

Measuring the Shape of Science

Morphological Indicators and Evolution of Research Fields

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Master Thesis Defense
July 2026

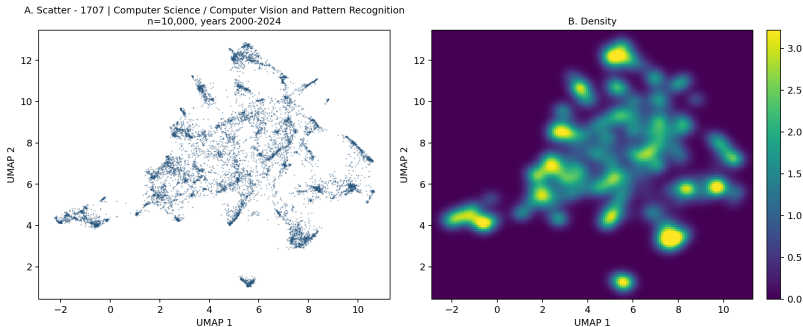
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Outline

- ① Problem and Measurement
- ② Static Morphology of Fields
- ③ Temporal Evolution
- ④ Relations Between Fields
- ⑤ Conclusions

A scientific field is a cloud of documents

A field label says what research is *called*, not how its papers are **arranged**.



One subfield, seen from above: 10,000 Computer Vision papers, placed by similarity of title and abstract.

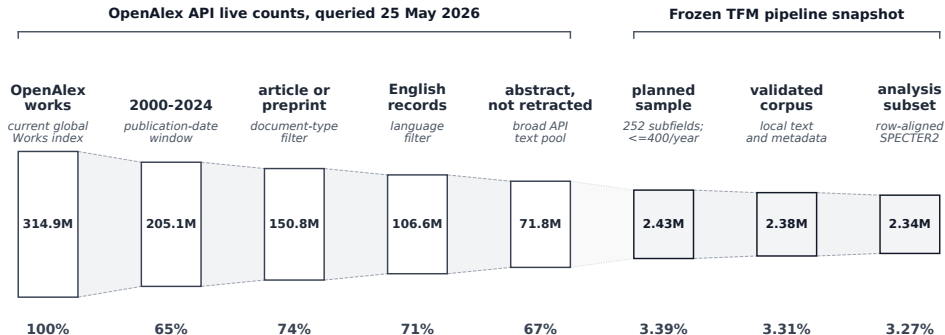
Central idea: treat each field as a **cloud of papers** and measure the **shape** of that cloud.

Three questions about the shape of science

- ① Can we **measure** the shape of a scientific field?
- ② Does that shape **differ** across sciences, and does it **change** over time?
- ③ Which fields are **moving closer together**, and which are **drifting apart**?

25 years of science, 2.3 million papers, one measuring stick.

A balanced OpenAlex corpus makes 241 subfields comparable



2,344,927 papers • **241** subfields • **26** fields • **4** domains

Subfields are the measuring unit; a field's shape is the average of its subfields.
Same yearly cap for every subfield: fields are compared on shape, not on publication volume.

Every paper becomes a point in the same space

- SPECTER2, a language model trained on scientific papers, reads each title and abstract and turns it into **768 numbers**: a point in space.
- Papers about similar research land **close together**; unrelated research lands far apart.
- A field is then a cloud of points, and the shape of that cloud **can be measured**.

One rule, applied everywhere

Every number in this thesis is computed in the full 768-dimensional space. Maps like the previous one are only for looking.

Eight numbers describe the shape of a field

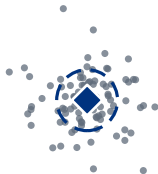
Family	Metrics	Question answered
Spread	Centroid median, IQR, P90	How far do papers spread out?
Local packing	kNN median, kNN CV	How tightly are neighbours packed?
Hubs	kNN in-degree Gini	Do a few papers dominate?
Dimensionality	PCA D_{80} , PCA entropy	How many directions are needed?

Eight numbers, kept separate

Each family captures a different kind of shape: a field can be broad but tightly packed, or compact but dominated by a few hub papers.

Spread: how far papers reach from the field centre

compact: papers stay near the centre



spread out: papers reach far



the real 241 subfields on this measure



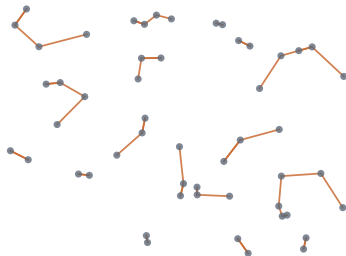
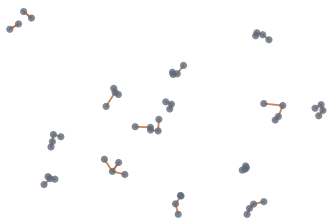
most compact: Applied Microbiology & Biotech.

most spread out: Nuclear & High-Energy Physics

Local packing: how close neighbouring papers sit

dense: each paper has neighbours right beside it

sparse: the nearest neighbour is far away



the real 241 subfields on this measure



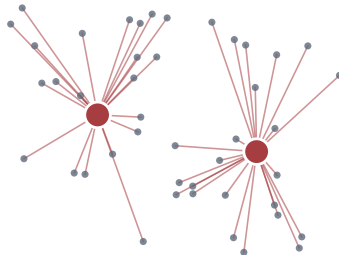
tightest neighbourhoods: Applied Microbiology & Biotech.

loosest neighbourhoods: Molecular Biology

Hubs: whether a few papers dominate

balanced: attention is spread evenly

hub-driven: many links point to the same few papers



the real 241 subfields on this measure

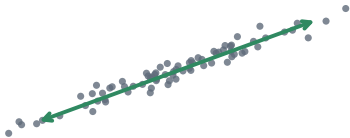


no dominant papers: Artificial Intelligence

a few papers dominate: General Materials Science

Dimensionality: how many directions the variation needs

simple: one direction explains almost everything complex: variation needs many directions



the real 241 subfields on this measure

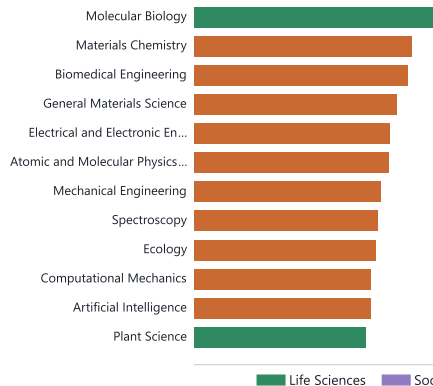


fewest directions: Business & Intl. Management

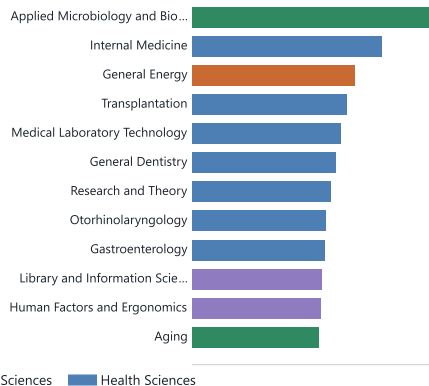
most directions: Classics

The loosest and tightest corners of science

the 12 loosest-packed subfields



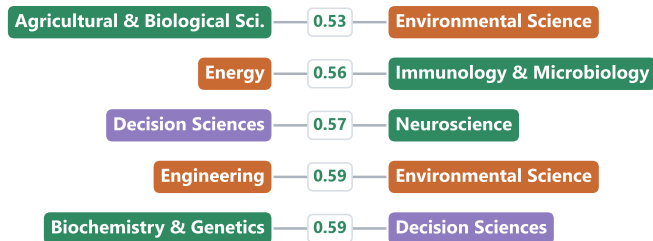
the 12 most tightly packed



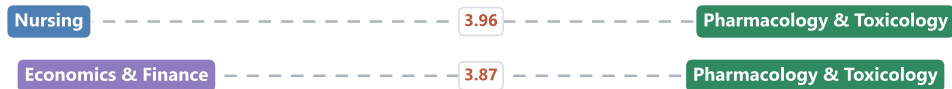
The loose end is Physical Sciences territory; the tight end belongs to medicine and health.

The most similar shapes ignore domain borders

MOST SIMILAR SHAPES

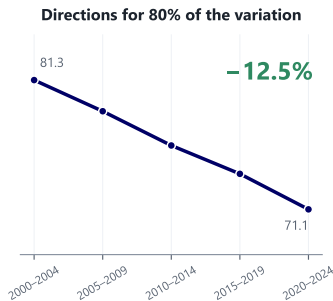
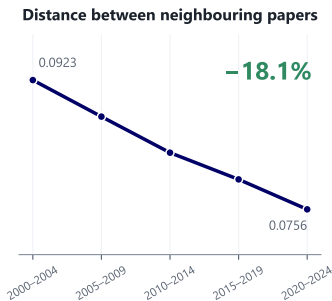
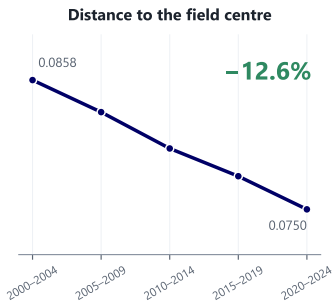


MOST DIFFERENT SHAPES



7 of the 10 closest pairs join fields from **different domains**.

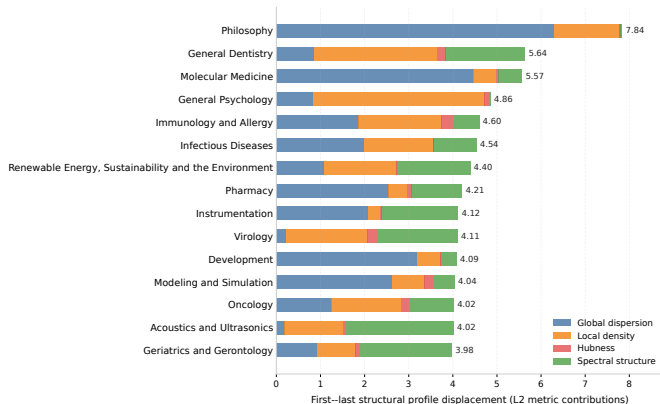
Science is packing tighter



Papers sit closer to their field's centre, closer to their neighbours, and vary along fewer directions.

And the tightening is **uneven**: some neighbourhoods shrink much faster than others.

Changing shape is not the same as moving

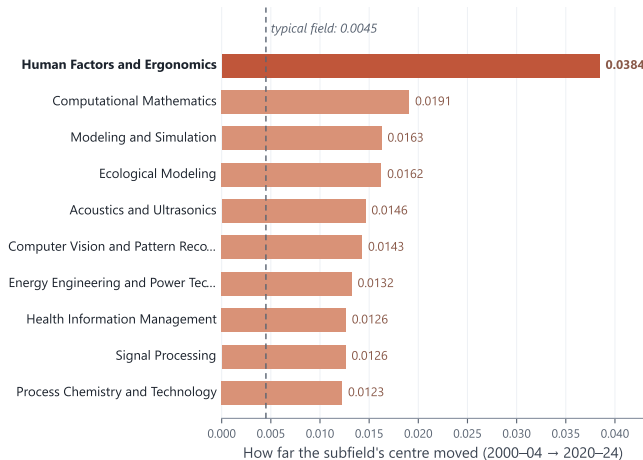


Biggest change in shape

Philosophy: its papers now spread far more *evenly* around the centre than in 2000.

The colours show which shape family drives each change: a different mix for every subfield.

Some fields travel, others stay put



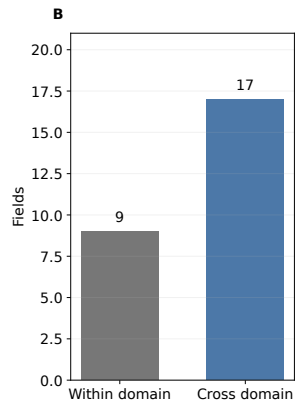
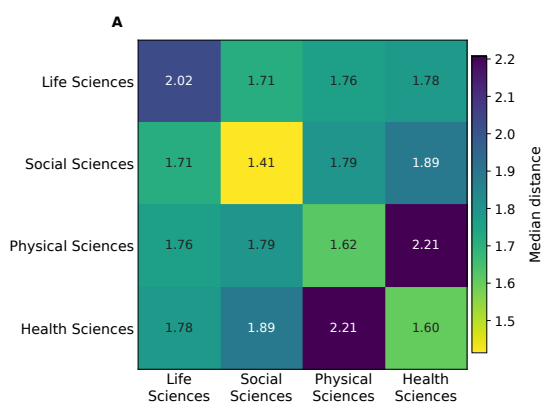
Biggest change of place

Human Factors and Ergonomics: its topics move $8.5\times$ further than the typical field.

The movers are applied, technology-driven areas. Mature theory (Mathematical Physics, Algebra, Geometry) barely moves in 25 years.

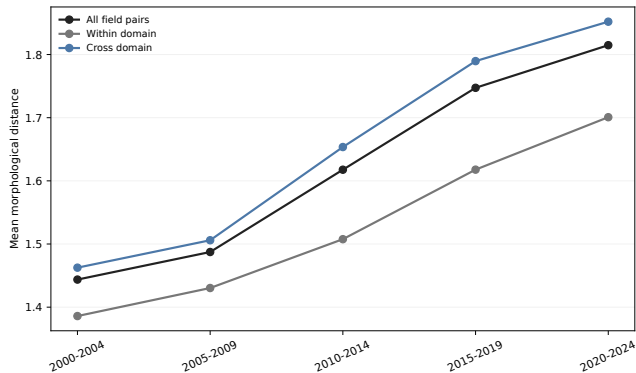
A field can rebuild itself without moving, or move without rebuilding.

17 of 26 fields look most like a field from another domain



Fields from the same domain are somewhat closer in shape (**1.62** vs. **1.86**), yet for **65%** of fields the most similar shape belongs to **another** domain.

Yet field shapes are drifting apart: +25.7% in twenty years



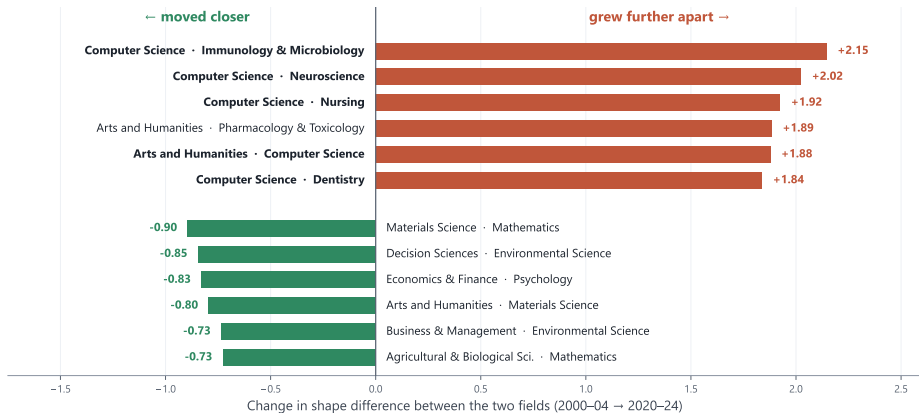
Average distance between fields

1.44 → 1.81

from 2000-2004 to 2020-2024

- **228** pairs of fields drift apart; only **97** move closer.
- The rise appears both inside domains and across them.

Computer Science is pulling away from everyone



Computer Science appears in **9 of the 10** biggest separations, while surprising couples such as Materials Science and Mathematics end up **50% closer**.

Main findings

Finding	Evidence
Papers pack tighter inside their fields	Distance between neighbouring papers –18%
Research varies along fewer directions	From 81 to 71 main directions
Field shapes drift apart	Average distance between fields +26%
Shape ignores the official map of science	17 of 26 fields match a field from another domain
Changing shape $\neq$ changing place	Philosophy reshapes; Human Factors moves

Science is packing tighter on the inside while its fields grow more different from each other.

What this thesis contributes

- A **new, reproducible way to measure the shape** of a scientific field, comparable across all of science and across 25 years.
- One framework that answers three questions at once: what shape a field has, **how it changes**, and **which fields move together**.
- Everything is open: the code, the data pipeline, and an **interactive viewer** anyone can explore.

And a clear road ahead

The same measuring stick can be re-run with other language models, full text, or citation networks, so every finding can be stress-tested.

Every field has a shape. Now we can measure it.

Thank you

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Interactive viewer: <https://aleetreny.github.io/Mapping-Science/>

Code and data: <https://github.com/aleetreny/Mapping-Science/>

Backup slides

Supporting material for questions

Backup: why SPECTER2

- The unit of analysis is a **scientific paper represented by its text**, so the model must embed documents, not sentences, topics, or citation nodes.
- SPECTER2 is trained on scientific papers and evaluated across document-level tasks in the **SciRepEval** benchmark.
- Its input is exactly what the corpus provides for every paper: **title plus abstract**.
- One fixed model gives one shared measurement space, so every comparison in the thesis is like for like.

Model-agnostic by design

The sampling plan, metrics, and aggregation never touch model internals: swapping the embedding model is a rerun, not a redesign.

Backup: measuring in 768 dimensions

Known concerns in high-dimensional spaces: distance concentration, anisotropy, and hubness (Aggarwal et al., 2001; Radovanović et al., 2010).

- 1 **Hubness is measured, not ignored.** The kNN in-degree Gini turns a known nuisance of these spaces into one of the eight signals.
- 2 **Nothing is interpreted in absolute terms.** Every claim compares subfields measured in the same space, with the same n and the same k , after robust scaling.
- 3 **Distortions apply identically to every subfield**, so orderings and comparisons remain meaningful even where the geometry is imperfect.

In short

The thesis does not claim the space is flat; it claims the same ruler is used everywhere.

Backup: the eight structural metrics

Metric	Definition	Interpretation
Centroid median	Median distance to subfield centroid	Typical spread around the centre
Centroid IQR	IQR of distances to centroid	Inequality of radial spread
Centroid P90	90th percentile of centroid distances	Outer-tail dispersion
kNN median	Median distance to $k = 15$ nearest neighbors	Typical local spacing
kNN CV	Coefficient of variation of kNN distances	Unevenness of local density
Hub Gini	Gini of directed kNN in-degree counts	Concentration around hub papers
PCA D_{80}	Components explaining 80% of variance	Effective dimensionality
PCA entropy	Entropy of the normalized PCA spectrum	Evenness of variance directions

Computed per subfield on L2-normalized SPECTER2 vectors with cosine distances. Net semantic displacement is measured separately as early-late centroid drift.

Backup: exact metric formulas

For a subfield s : centroid $c_s = \frac{1}{n_s} \sum_{i \in s} z_i$, cosine distance $r_i = d(z_i, c_s)$; directed k -NN sets $N_k(i)$ with $k = 15$; in-degrees $h_j = \sum_{i \in s} \mathbf{1}\{j \in N_k(i)\}$; PCA variances $\lambda_1, \dots, \lambda_p$ with $p_j = \lambda_j / \sum_{\ell} \lambda_{\ell}$.

Spread

$$\text{median}_i r_i, \quad Q_{75}(r) - Q_{25}(r), \quad Q_{90}(r)$$

Local packing $D_k = \{d(z_i, z_j) : j \in N_k(i)\}$

$$\text{median}(D_k), \quad \frac{\text{sd}(D_k)}{\text{mean}(D_k)}$$

Hubs in-degree Gini

$$G = \frac{2 \sum_{\ell} \ell h_{(\ell)}}{n_s \sum_{\ell} h_{(\ell)}} - \frac{n_s + 1}{n_s}$$

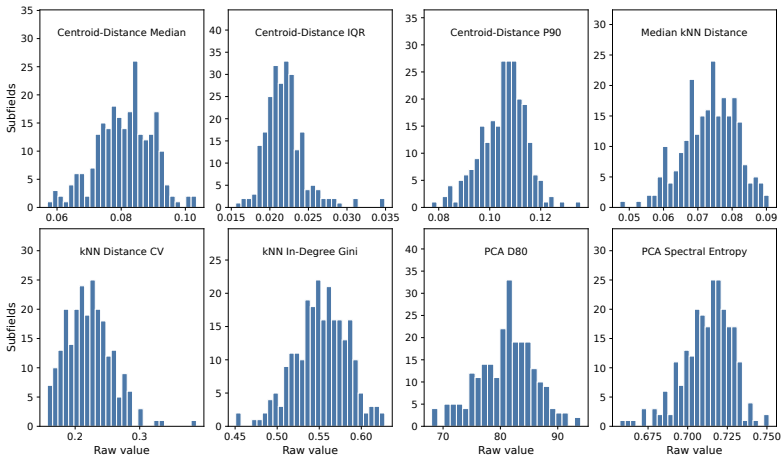
Dimensionality

$$D_{80} = \min \left\{ m : \frac{\sum_{j \leq m} \lambda_j}{\sum_j \lambda_j} \geq 0.80 \right\}$$

$$H = - \frac{\sum_j p_j \log p_j}{\log p}$$

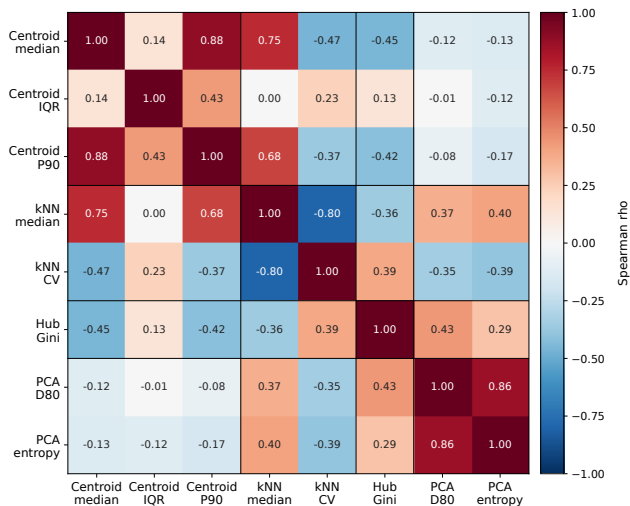
Separate indicator, semantic centroid drift: $\Delta_c(s) = d(c_s^{2000-04}, c_s^{2020-24})$.

Backup: raw metric distributions across subfields



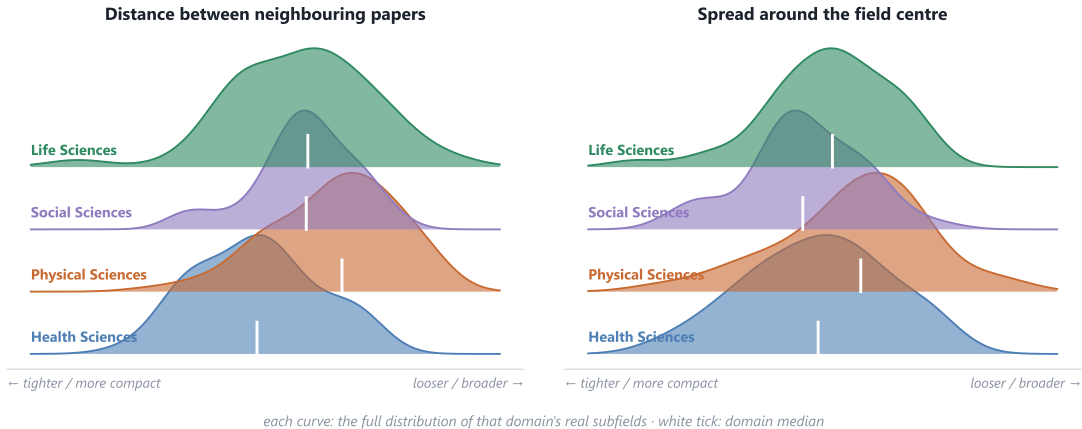
Several metrics are skewed or partly discrete; this motivates robust median/IQR scaling instead of z-scores.

Backup: metric families are related but not redundant

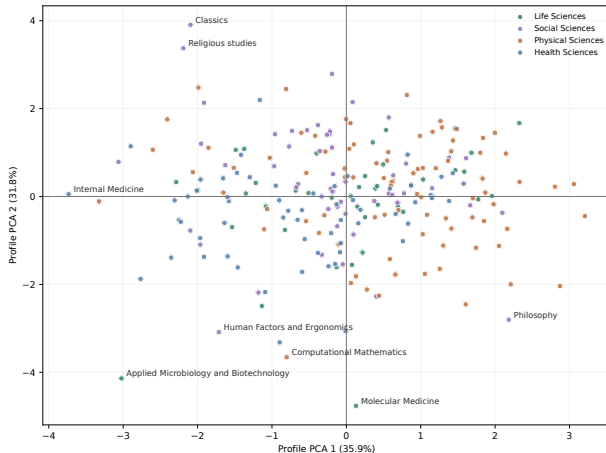


- Centroid median and P90:
 $\rho = 0.883$ (both radial spread).
- PCA D_{80} and entropy: $\rho = 0.856$ (both spectral).
- kNN median and kNN CV:
 $\rho = -0.798$ (closer neighborhoods pack less evenly).
- Cross-family correlations are weaker, supporting four separate families.

Backup: full distributions by domain

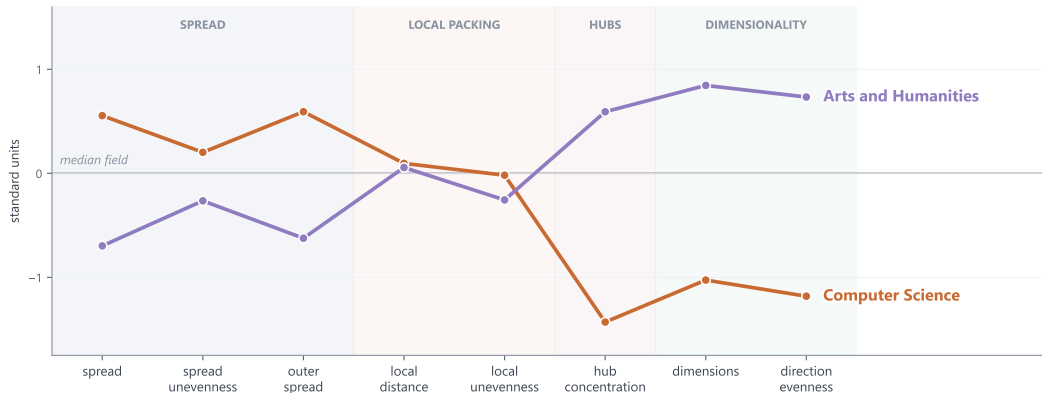


Backup: subfield profile space (PCA of standardized profiles)



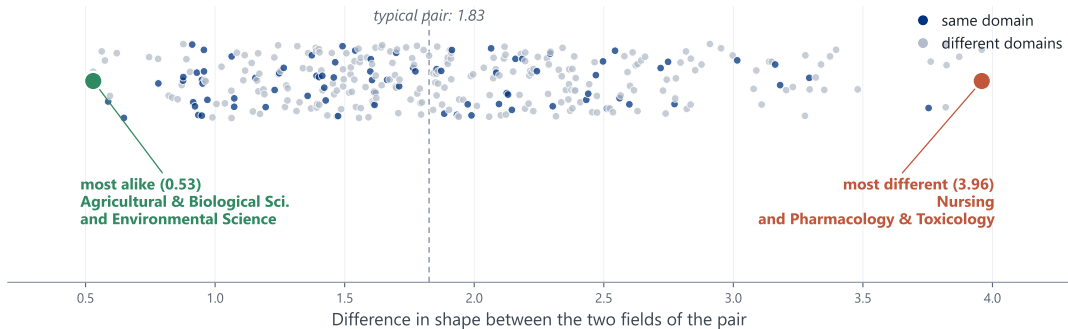
First two axes explain 35.9% and 31.8% of profile variance. Domains overlap near the centre; the informative cases are subfields at the edges.

Backup: two structural opposites, metric by metric



Real standardized field profiles. Computer Science is broad but hub-free and low-dimensional; Arts and Humanities is its mirror image.

Backup: all 325 field pairs by shape difference



Median pair difference 1.83; the closest pairs are mostly cross-domain.

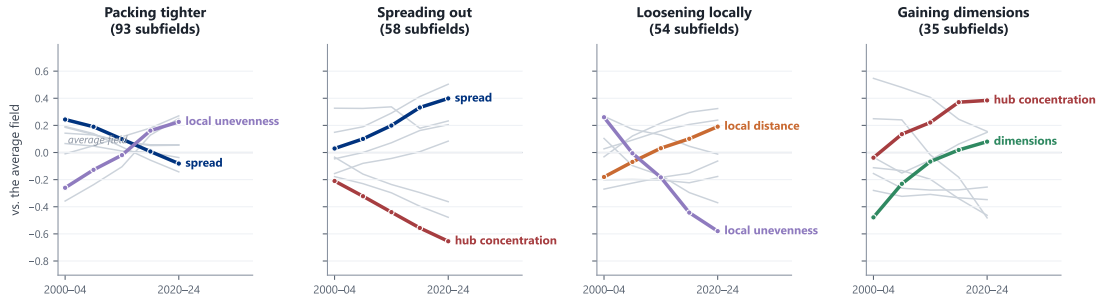
Backup: the kind of claims this thesis makes

Descriptive and comparative by design

How fields differ and change under one fixed corpus, taxonomy, model, and metric set. Not why, not how well, not how much they are worth.

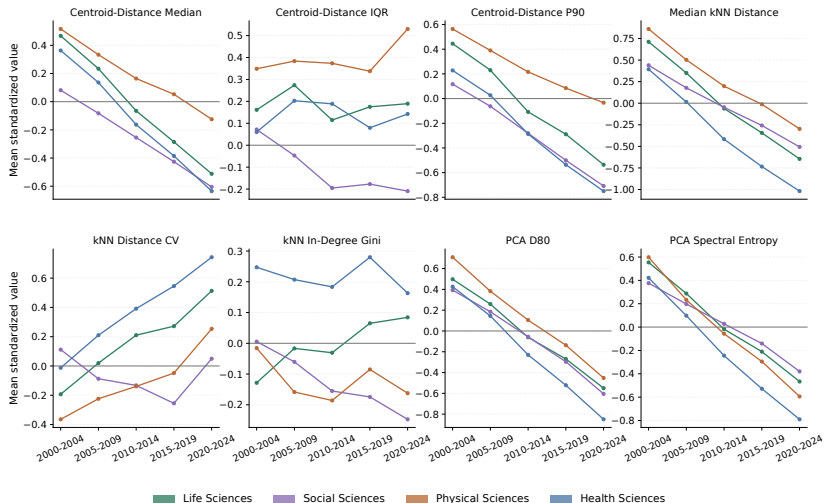
- Under that fixed design, the 241 subfields are the **whole population** of analysis units, not a sample from one; classical significance tests would answer a different question.
- Where variability does matter, it is quantified: the dynamic typology holds a **subsample bootstrap ARI of 0.725**.
- Robust median/IQR scaling keeps any single extreme subfield from driving a comparison.
- No claims about quality, impact, value, or causal mechanisms.

Backup: the four trajectory patterns

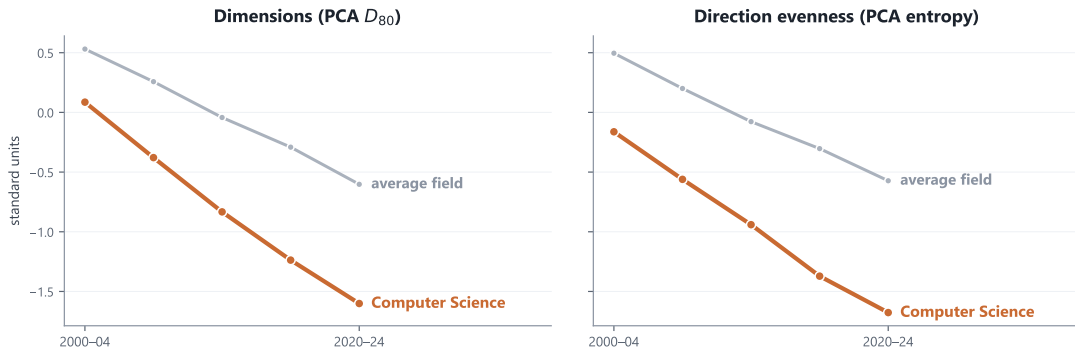


Cluster quality: silhouette 0.396 and subsample ARI 0.725 ($k = 4$), versus silhouette 0.174 for clustering on static profiles ($k = 5$).

Backup: domain-level temporal trajectories, all eight metrics

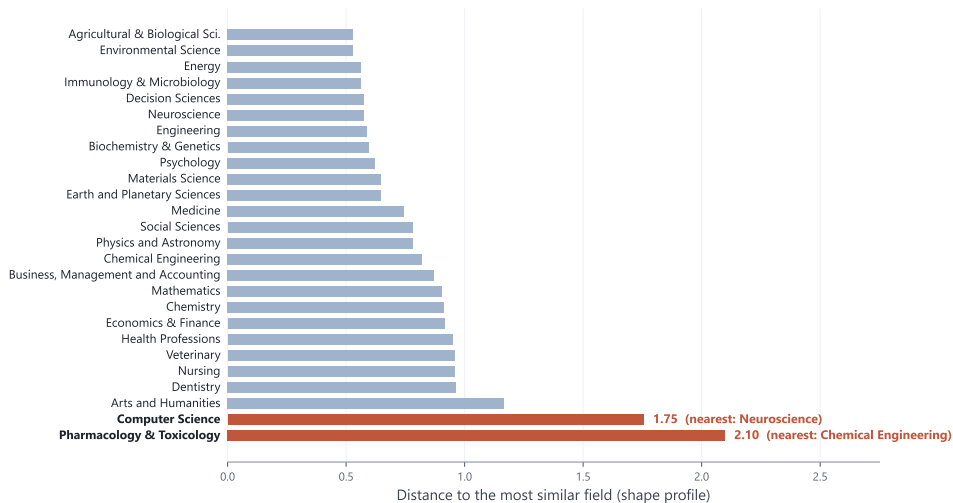


Backup: what is happening to Computer Science



Its variation collapses onto fewer, more dominant directions, much faster than science overall. That spectral collapse drives its nine largest divergences.

Backup: each field's nearest neighbour, and the two isolates



Backup: why every subfield has the same yearly cap

- kNN distances and PCA depend **mechanically** on sample size: more papers means closer neighbours and more components, regardless of any real structure.
- Without a cap, large fields would look dense and low-dimensional simply because they publish more.
- With the cap (400 papers per subfield and year, 10,000 over the period), every cloud is measured at a comparable n .
- Unused capacity is never moved to other years or subfields, and gaps are never imputed, so volume cannot re-enter the design.

The consequence

Fields are compared on shape, never on publication volume.

Backup: corpus validation

- Eligibility: English articles and preprints, 2000–2024, with sufficient title–abstract text.
- Checks passed: no duplicates, no out-of-period rows, no missing taxonomy or text fields, no disallowed types.
- Of 6,300 subfield–year cells, **5,493** reach the 400-work target; 807 sparse cells are coverage limits, not failures.
- Sparse capacity is **not** reallocated and gaps are **not** imputed, so volume cannot re-enter the design.
- Temporal matrix nearly complete: 1,204 of 1,205 subfield–window profiles (Computational Mathematics 2000–2004 excluded from first–last comparisons).

Backup: robustness and validation agenda

Representation

Rerun the identical pipeline with other scientific embedding models, full-text input, and multilingual corpora; one component swapped at a time.

Design sensitivity

Sweep k in the neighbour metrics; replace five-year blocks with annual and rolling windows; alternative taxonomies to the OpenAlex hierarchy.

External validation

Compare morphology with evidence deliberately excluded from its construction: citation networks, funding categories, and expert review of the extreme cases (Philosophy, Computer Science, Pharmacology).